

TECHBIZ SECURITY

CYBER SECURITY INTERNSHIP REPORT

Security Monitoring, Alert Investigation and Log Analysis Using LetsDefend

Submitted By

MANOJ

Role

Cyber Security Intern

Internship Organisation

TechBiz Security

Internship ID

TBS-INT-2026-518

Submitted Date

26 SEPT 2026

Task

Week 6 – Task 17 (EH)

1. Introduction

This report documents the practical activities carried out during Week 6, Task 17 (EH) of the Cyber Security Internship at TechBiz Security. The task involved hands-on security monitoring, alert investigation, vulnerability-related alert analysis, Windows authentication-event analysis, firewall and Remote Desktop Protocol (RDP) event investigation, and source IP investigation, all performed using the LetsDefend platform.

LetsDefend is a cybersecurity training environment designed to simulate a Security Operations Center (SOC) workflow. It provides a controlled lab setting containing pre-populated alerts, logs, and challenges that allow trainees to practice the day-to-day responsibilities of a security analyst without interacting with a real production system.

The screenshots included in this report provide practical evidence of the activities performed within this controlled training environment, including reviewing monitoring alerts, examining alert details, searching centralized logs, and analyzing Windows authentication and firewall events.

2. Objectives

- Understand security monitoring alerts.
- Investigate security alerts and their associated details.
- Examine vulnerability/exploitation-related alerts.
- Analyze Windows authentication events.
- Identify successful and failed login events.
- Investigate source and destination IP information.
- Analyze RDP-related firewall events.
- Understand how SIEM/log-management data can support incident investigation.
- Practice correlating multiple security events during an investigation.
- Document findings professionally.

3. Tools and Platform

Tool/Platform	Purpose
LetsDefend	Security monitoring, alert investigation and log analysis
Monitoring	Review security alerts
Log Management	Search and investigate security events
Event/Alert Details	Examine detailed security-event information

4. Lab / Training Environment

The practical investigation was performed within the LetsDefend cybersecurity training environment. This is a simulated SOC platform used for skill development and does not represent a live production network.

The screenshots captured during this task show the following categories of evidence:

- Monitoring alerts
- Alert details
- Log Management
- Windows authentication events

- Firewall events
- Source IP information
- Security-event metadata

Details such as operating system versions, network architecture, credentials, and machine configurations that are not visible in the screenshots are not included in this report.

5. Practical Investigation Overview

1. Step 1: Review the available LetsDefend challenges and security-analysis content.
2. Step 2: Review monitoring alerts and identify events requiring investigation.
3. Step 3: Open an alert and inspect detailed information such as severity, rule name, Event ID, alert type, hostname, IP address, process information, and MITRE ATT&CK mappings where available.
4. Step 4: Use Log Management to search through collected events.
5. Step 5: Investigate events associated with a destination IP address.
6. Step 6: Examine Windows Event ID 4624 and Event ID 4625 activity.
7. Step 7: Investigate firewall events involving RDP port 3389.
8. Step 8: Inspect the source IP associated with failed authentication activity.
9. Step 9: Correlate the available evidence and document security observations.

6. LetsDefend Challenge / Initial Evidence

The screenshot below shows the LetsDefend Challenge page containing a challenge titled "Suspicious Python Package." The challenge description indicates that forensic triage is used to investigate a potentially malicious Python package and its associated activities.

This screenshot reflects the challenge listing available on the platform. It does not by itself constitute evidence that the malicious package was personally analyzed; it establishes the training context in which the subsequent alert and log investigation activities in this report were carried out.

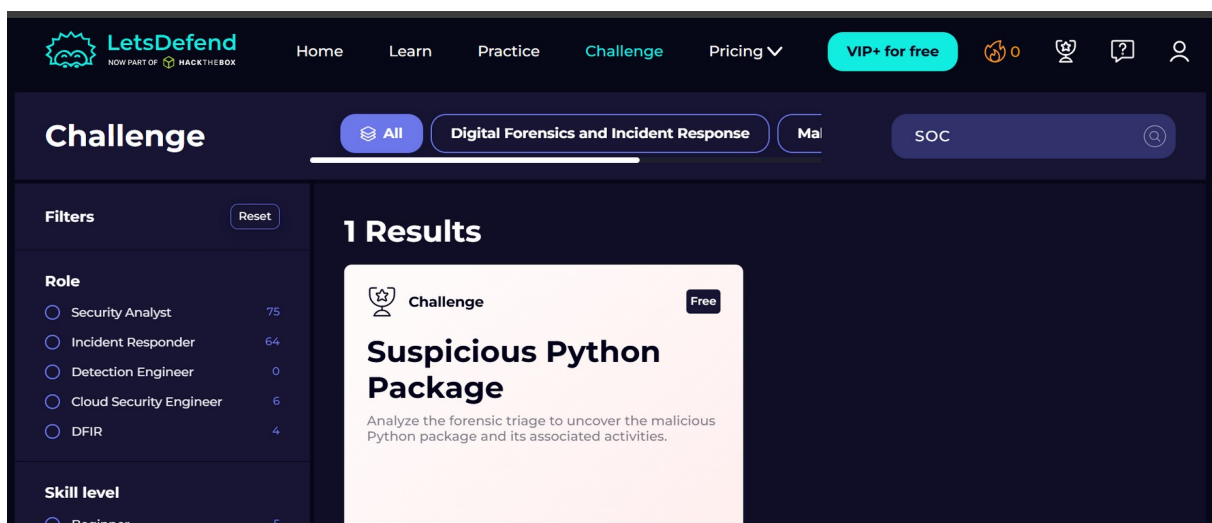


Figure 1: LetsDefend Challenge Page – Suspicious Python Package

7. Security Monitoring Alerts

The Monitoring interface displays multiple security alerts, each associated with a severity level, date, rule name, Event ID, and alert type. This dashboard represents the primary channel through which a security analyst becomes aware of events requiring review.

The following alerts are visible in the Monitoring dashboard, listed with their observed severity:

- SOC344 - EDR Tampering Attempt via EDR-Freeze (High)
- SOC343 - WinRAR Zero-Day Path Traversal Vulnerability (CVE-2025-8088) (Critical)
- SOC342 - CVE-2025-53770 SharePoint ToolShell Auth Bypass and RCE (Critical)
- SOC341 - Local Privilege Escalation via chroot CVE-2025-32463 (High)
- SOC340 - Apache Tomcat Serialized Payload RCE (CVE-2025-24813) (Critical)
- SOC339 - ZDI-CAN-25373 Windows Shortcut Exploit Detected (High)
- SOC338 - Lumma Stealer - DLL Side-Loading via Click Fix Phishing (Critical)

These entries represent alerts generated within the LetsDefend training platform. They were reviewed as part of the monitoring exercise; no claim is made that the underlying attacks were performed by the intern. The Monitoring interface helps an analyst prioritize investigation according to severity, rule name, Event ID, and alert type.

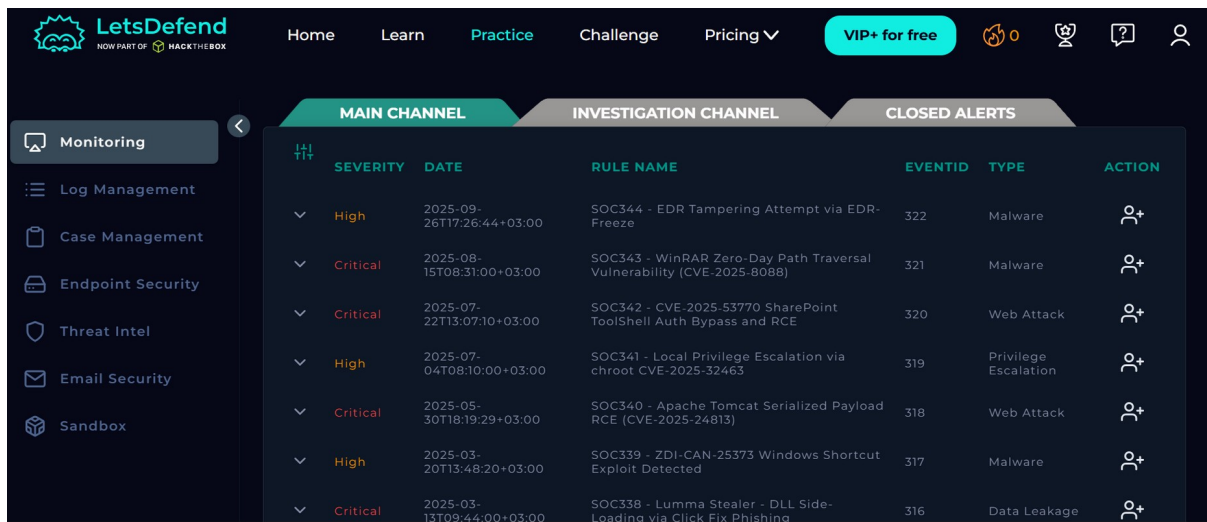


Figure 2: LetsDefend Monitoring Dashboard and Security Alerts

8. Detailed Alert Investigation – CVE-2024-49138

One alert from the Monitoring dashboard, "SOC335 - CVE-2024-49138 Exploitation Detected," was opened to examine its detailed information. This alert is labelled by the LetsDefend platform as an exploitation-detection alert; the evidence below reflects the alert data as investigated during the training exercise and does not indicate that CVE-2024-49138 was personally exploited.

Field	Value
Event ID	313
Rule	SOC335 - CVE-2024-49138 Exploitation Detected
Role	Security Analyst
Alert Type	Privilege Escalation

Field	Value
Difficulty	Medium
MITRE ATT&CK	T1059.001, T1055, T1068, T1548, T1110
Hostname	Victor
IP Address	172.16.17.207
Process ID	7640
Process Name	svhost.exe
Process Path	C:\temp\service_installer\svhost.exe
Process User	EC2AMAZ-ILGVOIN\LetsDefend
Command Line	\??\C:\Windows\system32\conhost.exe 0xffffffff -ForceV1
File Hash	b432dcf4a0f0b601b1d79848467137a5e25cab5a0b7b1224be9d3b6540122db9

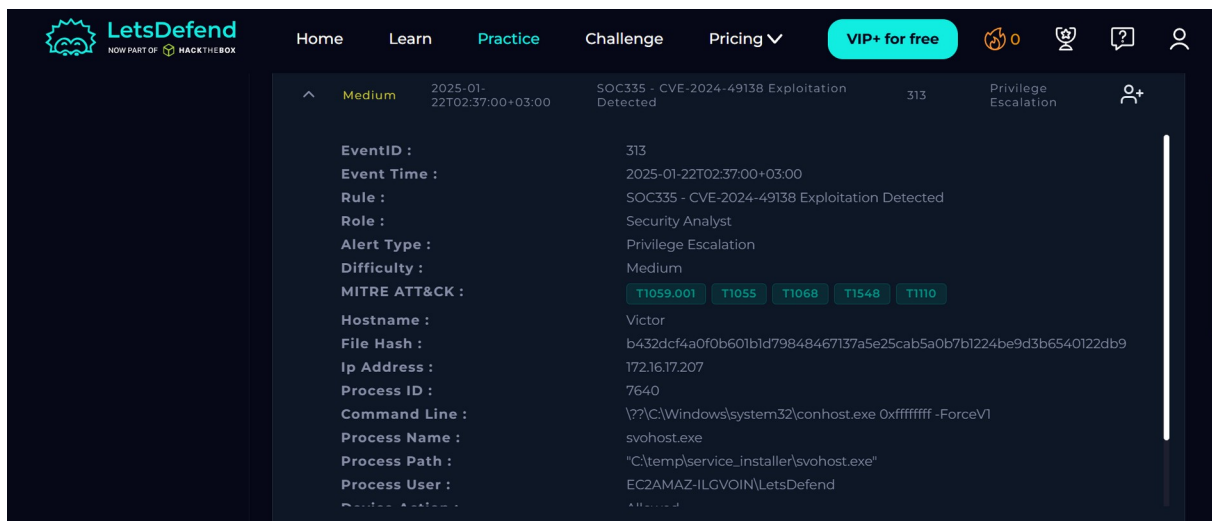


Figure 3: Detailed LetsDefend Alert – CVE-2024-49138 Exploitation Detected

Alert metadata such as process ID, process path, process user, hostname, IP address, and MITRE ATT&CK mappings is useful during incident investigation because it allows an analyst to identify the executable involved, the account context under which it ran, the host and network location of the activity, and the adversary techniques the behaviour maps to. This information supports triage decisions such as escalation, containment, and further evidence collection.

9. Log Management – Event Collection

The Log Management interface was used to review centrally collected security events. At the time of review, the interface reported 3907 events found, with event records containing timestamps, event type, source and destination addresses, and user/process information.

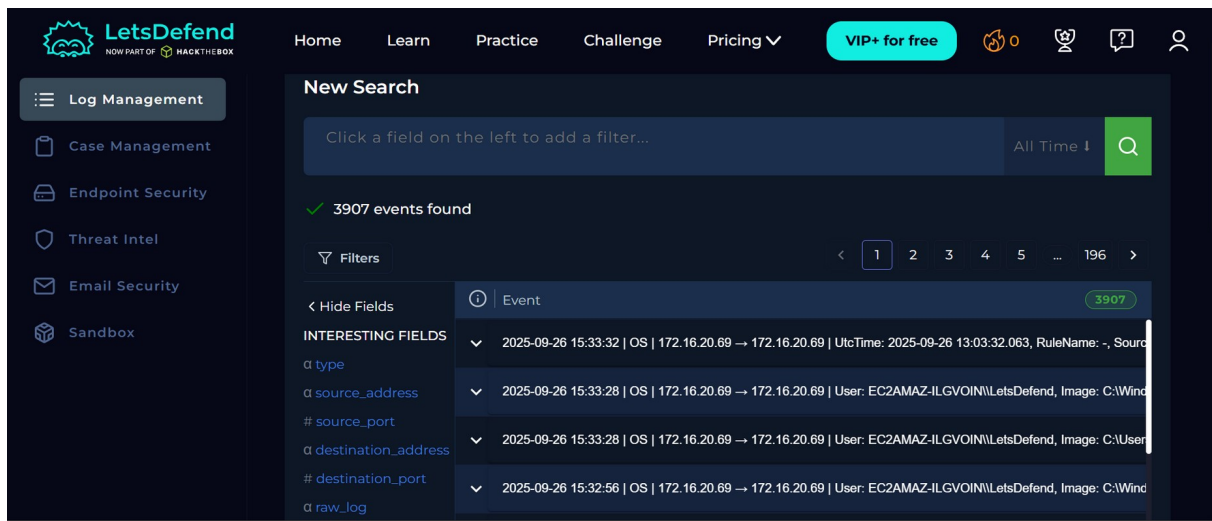


Figure 4: LetsDefend Log Management – Collected Security Events

Centralized log management allows a security analyst to search and investigate large volumes of security telemetry from a single interface, rather than reviewing logs on individual systems separately.

10. Destination IP Investigation

The Log Management search was filtered using the query: Destination Address contains "172.16.17.207". This search returned 10 events, including one Event ID 4624 entry and multiple Event ID 4625 entries, with a consistent source IP address of 185.107.56.141 and destination IP address of 172.16.17.207.

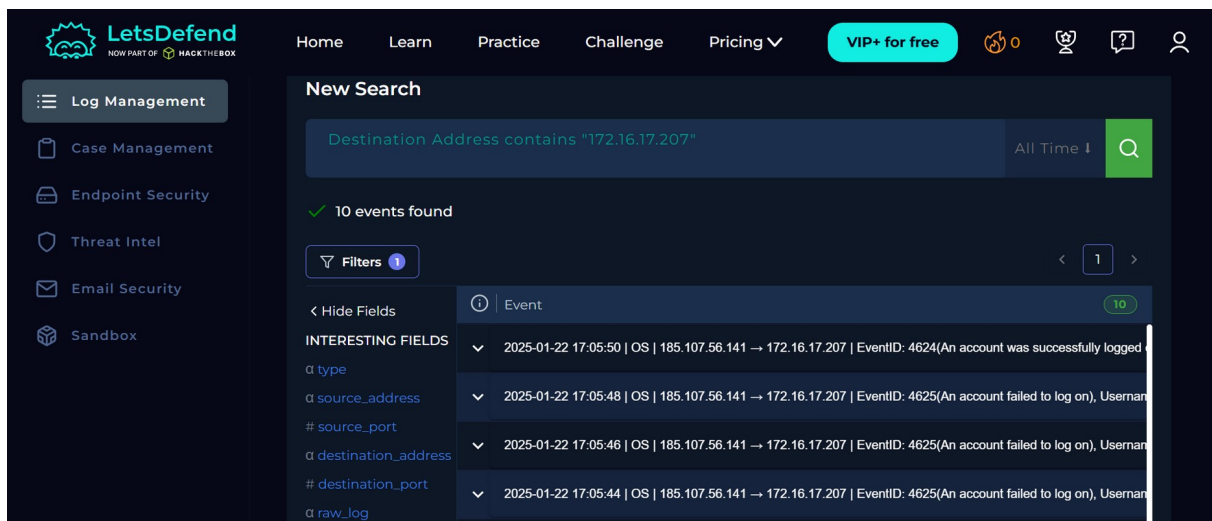


Figure 5: Search Results for Destination IP 172.16.17.207

Filtering by destination IP address allows an analyst to focus the investigation on activity associated with a specific host. The presence of failed logon events does not, by itself, establish that the associated source IP address is malicious; further correlation is required.

11. Windows Event ID 4624 – Successful Logon

Event ID 4624 represents a successful account logon in the displayed event data. The following values were observed:

Field	Value
Type	OS
Source Address	185.107.56.141
Source Port	0
Destination Address	172.16.17.207
Destination Port	0
Time	2025-01-22 17:05:50

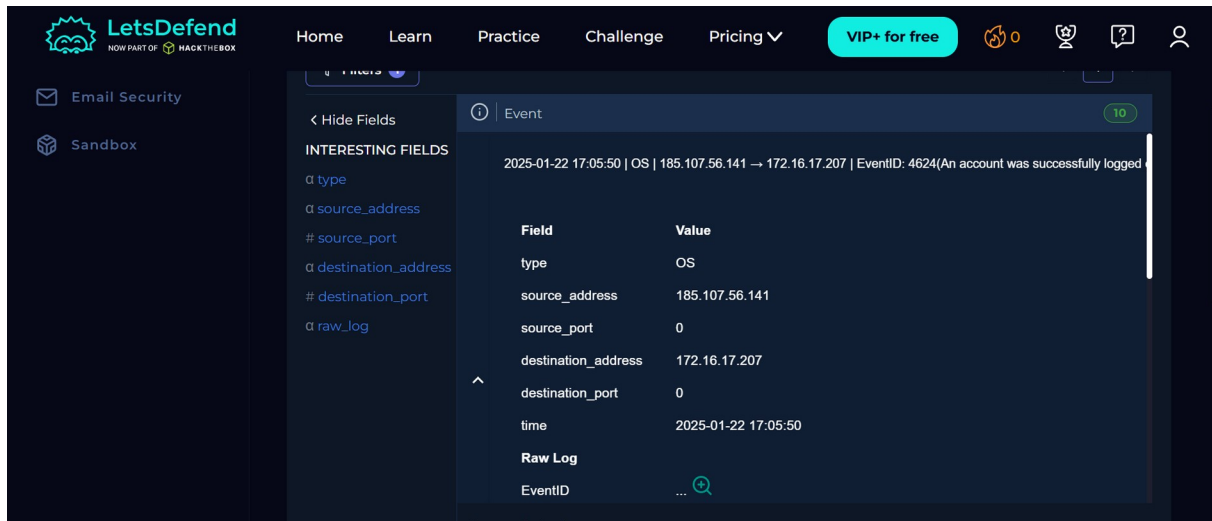


Figure 6: Windows Event ID 4624 – Successful Logon Details

A successful authentication event should be correlated with surrounding failed-login events, the source address, account information, and other available telemetry to determine whether the activity is expected or suspicious. The evidence reviewed does not establish that this successful login was unauthorized.

12. Windows Event ID 4625 – Failed Logon

Event ID 4625 represents a failed account logon in the displayed Windows event data. The following values were observed:

Field	Value
Type	OS
Source Address	185.107.56.141
Destination Address	172.16.17.207
Time	2025-01-22 17:05:44

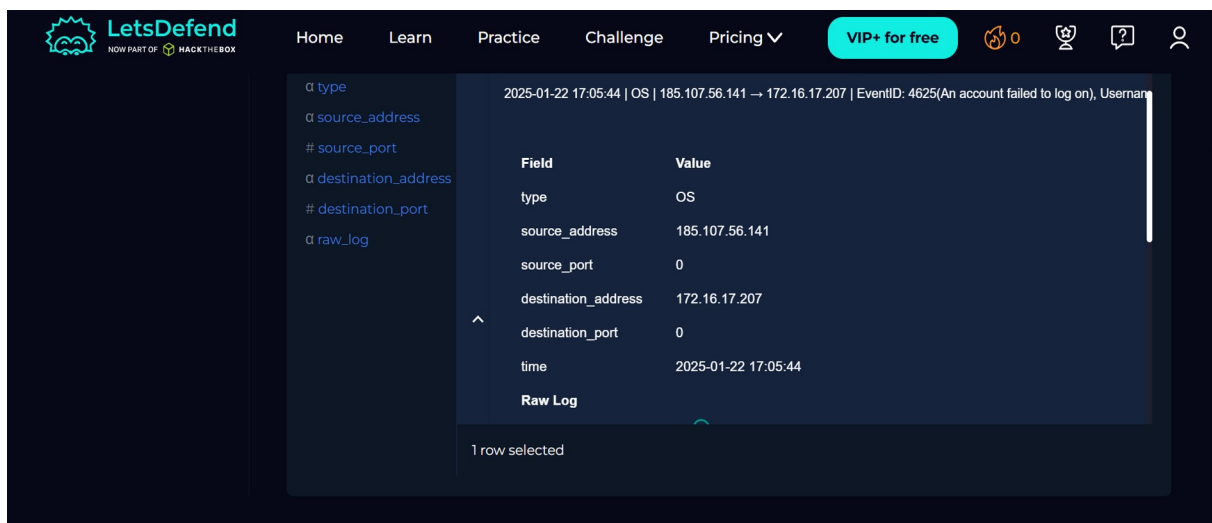


Figure 7: Windows Event ID 4625 – Failed Logon Details

Repeated Event ID 4625 events can be significant during authentication investigations because they may indicate incorrect credentials, user error, automated authentication attempts, or potentially malicious activity. Based on the evidence reviewed, the activity is not automatically classified as brute force without further supporting indicators.

13. Firewall and RDP Event Investigation

Firewall events associated with destination IP 172.16.17.207 and destination port 3389 were reviewed. Multiple entries were observed, each with a distinct source port and a source address beginning with 185.107.56...

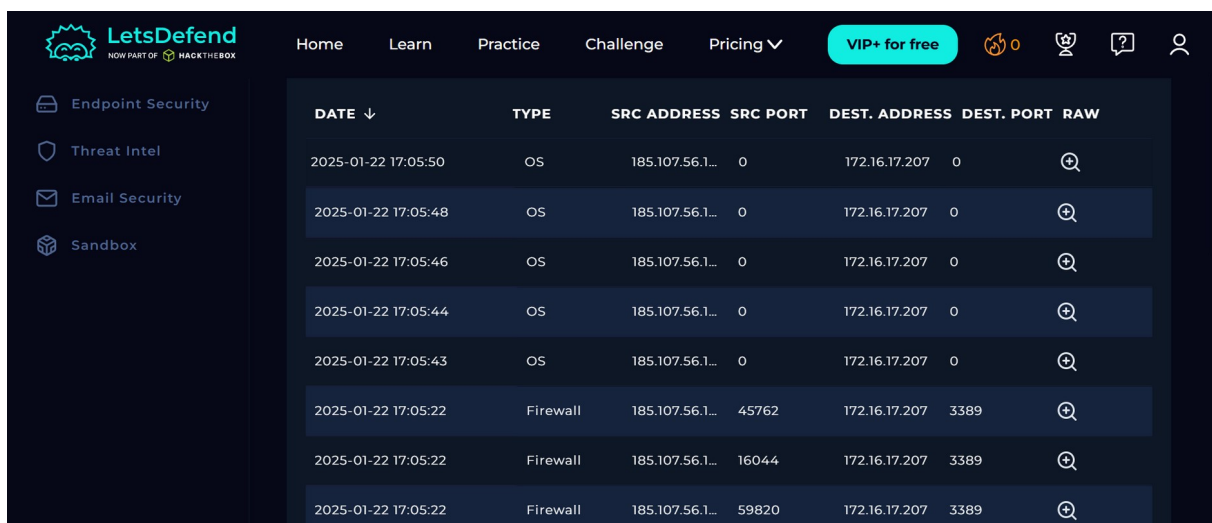


Figure 8: Firewall Events Associated with RDP Port 3389

TCP port 3389 is commonly associated with the Remote Desktop Protocol (RDP). RDP-related connection attempts should be reviewed in combination with authentication events to determine whether the connection activity was legitimate or suspicious. The screenshot does not provide evidence that an RDP session was successfully established.

14. Event ID 4625 Pop-up / Event Identification

Within the Log Management interface, the platform identifies the event as EventID: 4625 (An account failed to log on).

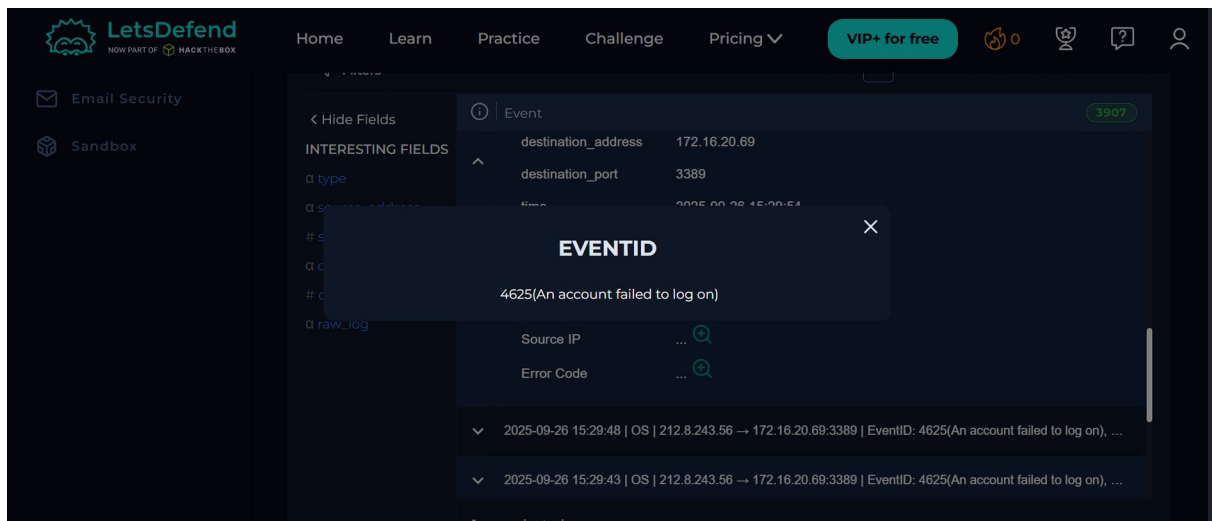


Figure 9: Event ID 4625 Identification

Event ID 4625 is significant in Windows security monitoring because it flags failed authentication attempts. This event supports authentication-related investigations when correlated with source IP, destination IP, timestamp, account information, other authentication events, and network/firewall activity.

15. Source IP Investigation

A separate event view within Log Management identifies the Source IP as 212.8.243.56. This value was observed in association with Event ID 4625 entries in that view.

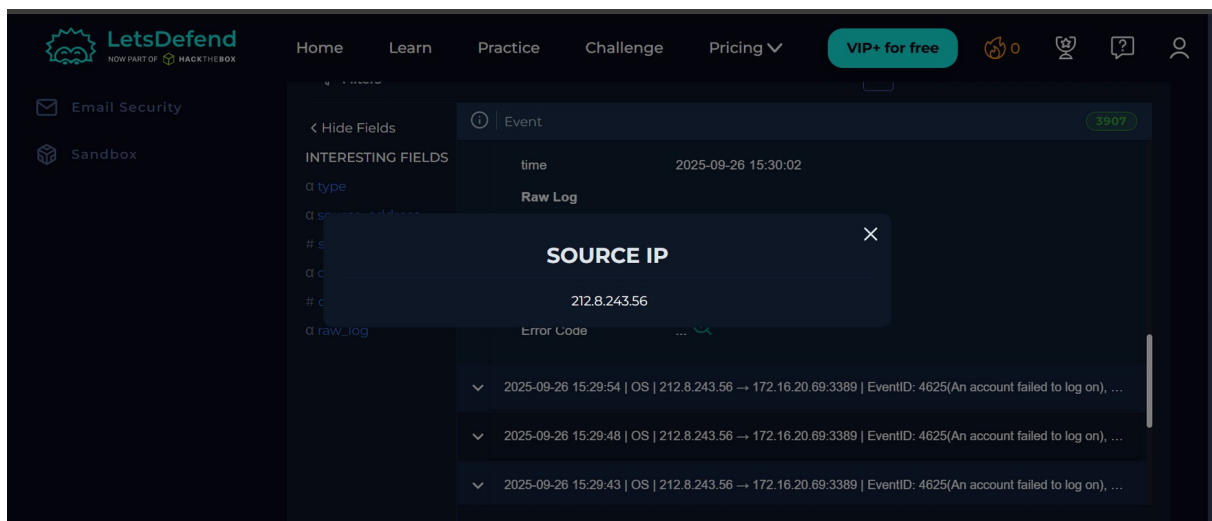


Figure 10: Source IP Identified During Event Investigation

This IP address is not labelled as malicious based on this screenshot alone. Source IP identification is useful for tracing the origin of network activity and for correlating multiple events during an investigation.

16. Event Correlation

The evidence gathered during this task forms a chain that can be used to build an investigation narrative. This chain is summarized below, using only the visible evidence:

- A. LetsDefend monitoring alerts provide initial security detections.

- B. A detailed alert provides additional context including Event ID, hostname, IP address, process information, and MITRE ATT&CK mappings.
- C. Log Management provides centralized access to security events.
- D. Searching for destination address 172.16.17.207 produces multiple events.
- E. Event ID 4625 indicates failed authentication activity.
- F. Event ID 4624 indicates a successful authentication event.
- G. Firewall events show activity involving destination port 3389.
- H. A separate investigation view identifies source IP 212.8.243.56.

Correlation across these different telemetry types can help a security analyst build an incident timeline. This evidence represents observed activity that requires further investigation; it does not conclusively prove a compromise, and the associated source IP addresses are not confirmed as attacker-controlled based solely on the screenshots reviewed. Such activity is best described as security-relevant and potentially suspicious, requiring correlation with additional telemetry.

17. Security Analyst Observations

Evidence	Observation	Security Relevance
Monitoring alerts	Multiple alerts with different severities are visible	Helps prioritize investigation
CVE-2024-49138 alert	Alert is classified as privilege escalation	Requires detailed investigation
Log Management	3907 events are available	Provides centralized event visibility
Destination IP search	10 events associated with 172.16.17.207	Helps narrow investigation scope
Event ID 4624	Successful logon observed	Authentication activity
Event ID 4625	Failed logon observed	Authentication failure requiring correlation
RDP port 3389	Firewall events observed	Indicates RDP-related network activity
Source IP	212.8.243.56 identified in an event view	Useful for event correlation

18. Investigation Timeline

The following timeline is based only on timestamps visible in the reviewed screenshots:

- 2025-01-22 17:05:22 – Firewall activity observed on destination port 3389.
- 2025-01-22 17:05:44 – Failed logon event (Event ID 4625) observed.
- 2025-01-22 17:05:46 – Failed logon event (Event ID 4625) observed.
- 2025-01-22 17:05:48 – Failed logon event (Event ID 4625) observed.
- 2025-01-22 17:05:50 – Successful logon event (Event ID 4624) observed.

This sequence shows multiple failed logon events followed by a successful logon event. This pattern should be investigated further through correlation with account information, source IP reputation, and surrounding network activity; no conclusion of compromise is drawn from the timeline alone.

19. Security Impact

- Repeated failed authentication attempts can increase the need for investigation.
- Successful authentication following failed attempts may require validation.
- RDP exposure can increase the importance of monitoring authentication activity.
- Vulnerability-related alerts require validation and investigation.
- Process-path and process-user information can help analysts identify suspicious execution.
- Centralized log management supports incident detection and investigation.

These points describe the general security relevance of the observed evidence. No breach or successful compromise is claimed based on the screenshots reviewed during this task.

20. Recommended Investigation and Defensive Actions

10. Validate whether the observed source IP addresses are authorized.
11. Review the affected account associated with Event ID 4624/4625.
12. Correlate authentication events with firewall and RDP logs.
13. Review additional events before and after the displayed timestamps.
14. Investigate unusual process paths and process execution.
15. Validate the legitimacy of the process user.
16. Review endpoint telemetry for suspicious execution.
17. Investigate vulnerability-related alerts and verify affected software versions.
18. Apply security updates and vendor patches where applicable.
19. Restrict unnecessary RDP exposure.
20. Apply network-level access controls for RDP.
21. Monitor repeated authentication failures.
22. Use SIEM correlation rules to connect authentication and network events.
23. Escalate confirmed malicious activity according to the organization's incident-response process.

21. MITRE ATT&CK Context

The following technique IDs are displayed by the LetsDefend alert examined in Section 8 and provide analytical context for that alert. These techniques were not independently verified by the intern beyond what is shown in the alert.

Technique ID	Visible in Alert
T1059.001	Yes
T1055	Yes
T1068	Yes
T1548	Yes
T1110	Yes

22. Skills Demonstrated

- Security monitoring
- Alert triage

- Log analysis
- Windows authentication-event analysis
- Event ID analysis
- IP address investigation
- Network-event correlation
- RDP traffic investigation
- Vulnerability-alert analysis
- MITRE ATT&CK awareness
- Security evidence documentation
- Incident investigation methodology

23. Evidence Summary

Figure	Screenshot	Evidence
Figure 1	01_LetsDefend_Suspicious_Python_Package_Challenge.png	Suspicious Python Package challenge
Figure 2	02_LetsDefend_Monitoring_Alerts.png.png	LetsDefend monitoring alerts
Figure 3	03_LetsDefend_Alert_Details_CVE_2024_49138.png	CVE-2024-49138 alert details
Figure 4	04_LetsDefend_Log_Management_Events.png	Log Management event collection
Figure 5	05_LetsDefend_Failed_Login_Events_4625.png	Destination IP event search
Figure 6	06_LetsDefend_EventID_4624_Details.png	Event ID 4624 details
Figure 7	07_LetsDefend_EventID_4625_Details.png	Event ID 4625 details
Figure 8	08_LetsDefend_Firewall_RDP_3389_Events.png	RDP port 3389 firewall events
Figure 9	09_LetsDefend_EventID_4625_Popup.png	Event ID 4625 identification
Figure 10	10_LetsDefend_Source_IP_212_8_243_56.png	Source IP investigation

24. Conclusion

This practical task provided hands-on exposure to security monitoring and investigation using the LetsDefend platform. The exercise demonstrated how an analyst can review alerts, examine alert metadata, search centralized logs, investigate authentication events, correlate successful and failed logons, examine RDP-related network events, identify source and destination IP information, use MITRE ATT&CK information as analytical context, and document security observations.

No claim is made that a real-world attack was successfully stopped or that a production system was compromised. All activities described in this report were performed within the LetsDefend training environment.

25. Final Learning Outcome

This exercise strengthened practical understanding of SOC/DFIR workflows, alert triage, log investigation, authentication-event analysis, network-event correlation, and security evidence documentation. It provided a structured introduction to how individual pieces of telemetry, such as alerts, authentication events, and firewall logs, are brought together during a security investigation.

26. References

- LetsDefend cybersecurity training platform
- Microsoft Windows Security Event ID documentation
- MITRE ATT&CK